and silicon is to be developed. Micro-mechanics, because—imagine this—tiny connectors actually *touching* individual neurons. Apart from the fact that it's all tiny, Sandini explains that bio-compatible materials need to be explored; in addition, a "micro-environment" suitable for the coexistence of the flesh and the wires needs to be devised.

Thinking from the ground up, then, we have more problems. Think of information coding. What kind of computer signal would a nerve be able to understand? How can—or should—we fire neurons? This involves deep study of the language that nerve cells use amongst themselves, and how that relates to the impulses we can generate.

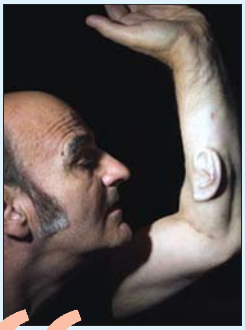
Now, if we're going to have a new kind of coded information, we're also going to need a new kind of information processing. Think about this: once an implant is in place, the tips of the nerves and the tips of the electrical interface cannot possibly be perfect. This means the processor would need to be tweaked a little to make for a better implant experience, and for that tweaking to happen, we need to make sense of the information actually flowing within the device.

We also need better and new types of batteries—as in everything digital today. Receivers of, say, artificial legs, can take their legs off at night and recharge them from the mains, but what about those with heart implants? Forget to recharge, and you pay for it with your life... of course, it's not quite that way, but we're talking about devices that can be powered by such things as body heat and limb movement. Like watches that don't require batteries, running on the energy generated by the movement of your forearm.

Then there is the problem of creating new sensors. Optical and acoustic sensors are reasonably acceptable today in terms of quality, but what about those used in smell and taste? Smell, as you know, occurs when molecules of a substance bombard sensors in the nose. We need to create devices that can actually translate molecular concentration into computer-friendly signals.

Links

http://tinyurl.com/285zhx
A cool robotic hand with ultra-cool fingers. Videos, too.
http://tinyurl.com/yuv677
"Physical Enhancement Page." Lots of links. Transhumanism and stuff.
http://tinyurl.com/2xg2ge
Excellent, long Wired article on The Desire To Be Wired.
http://tinyurl.com/yqkv5v
A company page. They call it "healthcare," and they talk about bionic stuff like store products. Don't skip the part about the biker.
http://tinyurl.com/yovamq
About the bionic products of Iceland-based Ossur, whom we mentioned.
http://tinyurl.com/2crlnt
Web site of Victhom, which develops bionic devices. Videos here too.
http://tinyurl.com/yvdmon
Artificial *everything*: researchers make kidneys from a cow's ear, and more.
http://tinyurl.com/27jdkl
The first completely artificial heart. Videos.



"The extra ear becomes a kind of Internet antenna"

Stelarc
Principal Research Fellow
Performance Arts Digital Research Unit
Nottingham Trent University

Endpoint: The Brain

In the ultimate analysis, to make implants work the way we want them to, we need to understand how the brain works. Grand dream, naturally, but little step by little step, that's the road ahead.

"Philosophers and psychologists have long noted that human perception has both analogue and digital characteristics," says Dr Sebastian Seung, a professor at MIT and co-author of the research report we'll soon talk about. "One neuron can make another active or inactive, but the intensity of the activity varies in a continuous way." In other words, apart from turning each other on and off, nerve cells also have analogue interactions. This is reflected in, say, visual perception: we make spontaneous yes / no distinctions like whether it is light or dark outside, but we also see shades of grey when we need to.

In what seems a breakthrough idea to us—if not a breakthrough implementation—researchers at MIT have, using this concept, built a circuit of artificial neurons with hybrid analogue / digital interconnections. Details are sketchy, but the creators believe such circuits can—after refinement—help process auditory and visual information for robots. They could also process feedback information from bionic implants: retinal chips—as in the artificial eye we talked about earlier—could at some point be pre-processors feeding into such circuits, for example.

What is special about the circuit, the way we see it, is that it mimics the brain. It's a top-level idea, and it's not been seen in action yet, but think about it: since the brain is the final processor of information from and to any source in the body—biological or artificial—it's natural that the circuits that interface with it should behave like it.

Such is the road ahead: "Continued research in neuroscience and bioengineering will no doubt lead to improvements in man-machine interfaces and functional replacements, but it will be a long, hard road filled with many failures and a few successes." That's William Jenkins, Vice President for Development at Scientific Learning, which makes products that develop learning and communication skills.

Endnote

So bionics helps the blind see, the wheelchaired walk, the deaf hear, and more. It's interesting that some people look upon the field with a kinky sort of fascination: going beyond need, as in patients who need prostheses, they *want* implants. What if we could have more than two arms, Hindu-god-style? When will such things be socially acceptable?

We're veering away again to the realm of cyborgs, androids, and other such interesting creatures, but then, bionics does straddle the middle ground between corrective surgery and all this.

Like we said, it's not just a question of our current methods getting better: there are fundamental issues to be addressed. Again, like we said, there needs to be a better understanding of brain function for bionics to take off.

We cannot end on a predictive note, saying when those legs will be on the shelves: Jenkins is probably right. It will be a long, hard road. ⊠

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